

MCENTIRE JNGB, SC, USA

WMO: 723105

Lat: 33.921N

Lon: 80.801W

Elev: 77

StdP: 100.40

Time Zone: -5.00 (NAE)

Period: 08-19

WBAN: 03858

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-5.6	-3.5	-15.5	1.0	-0.2	-12.8	1.3	1.5	10.0	12.0	8.9	12.2	1.5	310	0.382

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.5	35.8	24.3	34.6	24.2	33.4	24.0	26.2	32.1	25.7	31.4	25.2	30.7	3.6	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
24.9	20.1	27.6	24.3	19.4	27.3	23.9	18.9	27.0	81.2	32.3	79.1	31.5	77.3	31.0	28.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.2	7.1	5.7	-8.9	37.4	2.2	1.8	-10.5	38.7	-11.8	39.7	-13.0	40.7	-14.6	42.0		
			-9.9	27.0	2.0	0.7	-11.3	27.5	-12.5	27.9	-13.7	28.3	-15.1	28.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	17.8	7.1	9.8	12.9	17.6	22.0	26.0	27.3	26.6	24.2	18.1	11.7	9.7
	DBStd	8.14	5.65	5.48	5.27	3.87	3.16	2.17	1.71	1.87	2.78	4.30	4.44	5.15
	HDD10.0	331	126	67	32	2	0	0	0	0	0	2	31	72
	HDD18.3	1371	348	242	180	59	9	0	0	0	1	57	203	272
	CDD10.0	3174	37	62	122	229	373	479	536	515	424	253	82	62
	CDD18.3	1173	1	3	12	37	124	229	277	257	176	50	4	4
	CDH23.3	11394	9	36	153	438	1203	2316	2829	2389	1518	462	31	9
	CDH26.7	5042	0	2	29	108	466	1117	1406	1117	669	125	2	0
Wind	WSAvg	2.3	2.5	2.7	2.8	2.7	2.4	2.1	2.0	1.8	2.0	2.1	2.1	2.3
Precipitation	PrecAvg	1073	53	86	83	68	87	125	125	124	90	73	64	88
	PrecMax	1360	101	136	177	161	163	201	317	229	223	277	131	215
	PrecMin	782	27	47	15	21	17	46	48	55	4	4	7	25
	PrecStd	179	22	25	38	32	34	37	73	46	52	63	36	52
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	24.8	26.8	29.4	30.9	35.6	37.4	37.4	36.3	35.2	32.0	26.5	24.7
		MCWB	17.7	18.2	17.7	19.7	21.9	24.1	24.3	24.9	22.8	23.8	19.7	19.5
	2%	DB	21.4	24.3	27.3	29.5	32.7	35.2	35.8	34.7	33.6	29.7	23.9	22.5
		MCWB	15.5	17.0	17.6	19.2	21.4	24.4	24.7	24.5	22.9	21.4	17.2	18.5
	5%	DB	19.1	22.0	25.0	27.9	31.0	33.7	34.6	33.4	32.3	27.8	21.9	20.5
		MCWB	14.9	15.5	16.2	18.9	21.5	24.0	24.8	24.6	22.9	20.2	16.0	17.2
	10%	DB	16.7	19.5	22.4	26.0	29.3	32.3	33.1	32.1	30.7	25.9	20.0	18.6
		MCWB	13.2	14.1	15.1	18.0	21.0	23.7	24.7	24.2	22.6	19.5	15.8	16.3
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	19.4	19.7	19.9	22.9	24.3	26.3	26.9	26.8	25.6	25.1	20.9	21.4
		MCDB	21.0	23.7	24.8	27.1	28.8	33.0	33.8	32.3	30.5	27.6	23.0	22.8
	2%	WB	17.8	18.4	19.0	21.7	23.5	25.6	26.2	26.1	24.9	24.0	19.6	19.9
		MCDB	20.6	22.7	24.0	25.9	28.3	32.2	32.4	31.6	29.8	27.0	21.8	21.5
	5%	WB	16.1	17.0	18.0	20.6	22.8	25.1	25.7	25.6	24.2	22.9	18.5	18.2
		MCDB	18.0	20.5	22.7	24.9	27.7	31.5	31.7	30.8	28.9	25.5	20.5	19.6
	10%	WB	14.0	15.4	16.7	19.5	22.2	24.5	25.2	25.2	23.7	21.3	16.8	16.5
		MCDB	15.8	18.3	20.7	23.7	26.9	30.3	31.0	30.0	28.0	24.4	19.4	18.3
Mean Daily Temperature Range	5% DB	MDBR	12.3	13.0	13.3	13.8	12.7	12.0	11.5	10.8	11.4	13.3	13.4	11.6
		MCDBR	14.6	15.5	16.1	16.0	15.2	13.8	13.5	12.8	13.8	14.6	15.3	12.9
		MCWBR	9.3	9.0	7.5	7.1	5.7	4.7	4.2	4.2	4.6	6.8	9.0	8.7
	5% WB	MCDBR	11.1	13.3	13.4	12.4	11.9	12.2	11.6	11.1	11.4	11.1	11.2	10.3
		MCWBR	8.5	9.1	6.8	6.5	4.8	4.7	4.2	4.0	4.2	6.2	8.2	8.4
Clear-Sky Solar Irradiance	taub	0.323	0.332	0.369	0.411	0.476	0.510	0.555	0.525	0.441	0.400	0.349	0.319	
	taud	2.491	2.463	2.384	2.261	2.124	2.072	1.969	2.045	2.265	2.346	2.450	2.539	
	Ebn at Noon	889	918	905	878	821	789	751	767	821	829	848	874	
	Edh at Noon	86	98	114	135	157	165	182	166	128	109	89	78	
All-Sky Solar Radiation	RadAvg	2.78	3.41	4.49	5.66	6.15	6.39	6.09	5.53	4.76	3.99	3.06	2.43	
	RadStd	0.15	0.36	0.34	0.41	0.48	0.45	0.39	0.28	0.41	0.48	0.24	0.22	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (62 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	+192	+135

Nomenclature: See separate page